WORKED SOLUTION

STEP III 1993, Question 9

Pure • Sums of Squares Inequalities

STEP III 1993, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q9

The key idea

Every inequality here is a disguised statement that a sum of squares is non-negative. Part (i) is the simplest case, $(a_1 - a_2)^2 \geq 0$, and the pattern for the later parts is to add up $(a_i - a_j)^2$ over *all* pairs and count how often each square and each product appears. Once the general version (iv) is in hand, (v) follows by expanding $(\sum a_i)^2$ into squares plus cross terms.

Throughout we write

$$S = \sum_{i=1}^{n} a_i^2, \qquad P = \sum_{i<j} a_i a_j,$$

where the second sum runs over the $\binom{n}{2}$ pairs with $1 \leq i < j \leq n$.

Solution

Part (i)

Since the square of a real number is non-negative,

$$0 \leq (a_1 - a_2)^2 = a_1^2 + a_2^2 - 2a_1a_2,$$

so $a_1^2 + a_2^2 \geq 2a_1a_2$, with equality exactly when $a_1 = a_2$.

Part (ii)

Apply (i) to each of the three pairs and add:

$$\begin{aligned} a_2^2 + a_3^2 &\geq 2a_2a_3, \\ a_3^2 + a_1^2 &\geq 2a_3a_1, \\ a_1^2 + a_2^2 &\geq 2a_1a_2. \end{aligned}$$

Each square appears in exactly two of these lines, so adding gives

$$2(a_1^2 + a_2^2 + a_3^2) \geq 2(a_2a_3 + a_3a_1 + a_1a_2),$$

and dividing by 2 gives the result. Equivalently, $a_1^2 + a_2^2 + a_3^2 - (a_2a_3 + a_3a_1 + a_1a_2) = \frac{1}{2}\left[(a_1 - a_2)^2 + (a_2 - a_3)^2 + (a_3 - a_1)^2\right] \geq 0$, with equality exactly when $a_1 = a_2 = a_3$.

Part (iii)

Now there are six pairs (i, j) with $1 \leq i < j \leq 4$. Apply (i) to each:

$$\sum_{1 \leq i < j \leq 4} (a_i^2 + a_j^2) \;\geq\; 2 \sum_{1 \leq i < j \leq 4} a_i a_j.$$

On the left, how often does a_1^2 appear? Once for every pair containing the index 1, that is, with each of the three other indices. The same holds for every a_k^2, so the left-hand side is $3(a_1^2 + a_2^2 + a_3^2 + a_4^2)$, and we have

$$3(a_1^2 + a_2^2 + a_3^2 + a_4^2) \geq 2(a_1a_2 + a_1a_3 + a_1a_4 + a_2a_3 + a_2a_4 + a_3a_4),$$

as required.

Part (iv)

The pattern in (i)–(iii) is that the coefficient on the left is one less than the number of variables: 1, then 2 (part (ii) multiplied by 2), then 3.

Claim. For any $n \geq 2$ and real numbers $a_1, \ldots, a_n$,

$$(n-1) \sum_{i=1}^{n} a_i^2 \;\geq\; 2 \sum_{1 \leq i < j \leq n} a_i a_j, \qquad \text{that is,} \qquad (n-1)S \geq 2P,$$

with equality if and only if $a_1 = a_2 = \cdots = a_n$.

> **?** **Ask yourself** When we add up $a_i^2 + a_j^2$ over every pair $i < j$, how many times does a particular a_k^2 turn up?

Proof. Consider the sum of the squared differences over all pairs:

$$D = \sum_{1 \leq i < j \leq n} (a_i - a_j)^2 = \sum_{1 \leq i < j \leq n} \left(a_i^2 + a_j^2\right) - 2 \sum_{1 \leq i < j \leq n} a_i a_j.$$

The index k belongs to exactly $n - 1$ pairs (it is paired with each of the other $n - 1$ indices), so a_k^2 appears $n - 1$ times in the first sum. Hence

$$D = (n-1)S - 2P.$$

Each term of D is a real square, so $D \geq 0$, which gives $(n-1)S \geq 2P$. Equality holds if and only if every $(a_i - a_j)^2$ is zero, that is, if and only if all the a_i are equal. ■

Putting $n = 2, 3, 4$ recovers (i), twice (ii), and (iii).

Part (v)

Here we need $n \geq 2$ so that $n-1 > 0$. Expanding the square of the sum, every square appears once and every product $a_i a_j$ with $i \neq j$ appears twice (as $a_i a_j$ and $a_j a_i$), so

$$\left(\sum_{i=1}^{n} a_i\right)^2 = S + 2P.$$

Ask yourself How can (iv), which bounds S, help with a bound that only mentions P?

Since $n-1 > 0$, part (iv) can be divided by $n-1$ without reversing the inequality, giving $S \geq \dfrac{2P}{n-1}$. Substituting this lower bound for S:

$$\begin{aligned}\left(\sum_{i=1}^{n} a_i\right)^2 &= S + 2P \\ &\geq \frac{2P}{n-1} + 2P \\ &= \frac{2P + 2(n-1)P}{n-1} \\ &= \frac{2n}{n-1} \sum_{1 \leq i < j \leq n} a_i a_j,\end{aligned}$$

which is the required inequality. Equality holds exactly when it holds in (iv), that is, when all the a_i are equal.

Note that we never needed P to be positive: the argument only adds $2P$ to both sides of $S \geq 2P/(n-1)$, which is valid whatever the sign of P.

Checks

- All equal: with every $a_i = c$, we get $S = nc^2$ and $P = \binom{n}{2}c^2 = \frac{1}{2}n(n-1)c^2$, so $(n-1)S = n(n-1)c^2 = 2P$, and in (v) both sides equal n^2c^2. Both inequalities are tight, as they should be.
- A numerical test of (iii): $(a_1, a_2, a_3, a_4) = (1, 2, -3, 5)$ gives $S = 39$ and $P = -7$, so $3S = 117 \geq -14 = 2P$; and in (v), $(\sum a_i)^2 = 25 \geq \frac{8}{3}(-7) = -\frac{56}{3}$. (Also $S + 2P = 39 - 14 = 25$, as it should be.)
- We also tested the identity $D = (n-1)S - 2P$ and both inequalities exactly, with rational arithmetic, on 20 000 random lists of length 2 to 9, plus lists clustered very close to equality. No counterexamples appeared.

Common slips

- Miscounting: each a_k^2 appears $n-1$ times in $\sum_{i<j}(a_i^2+a_j^2)$, not n times and not $\binom{n}{2}$ times.
- Writing $\left(\sum a_i\right)^2 = S+P$: each cross term appears twice in the expansion.
- Dividing (iv) by $n-1$ without noting that $n-1>0$, or forgetting to state the equality case.

Beyond STEP

The quantity $D=(n-1)S-2P$ is n times the sum of squared deviations from the mean, $n\sum(a_i-\bar{a})^2$, so (iv) is really the statement that a variance is non-negative. Part (v) can also be read as the Cauchy–Schwarz inequality $\left(\sum a_i\cdot 1\right)^2 \le n\sum a_i^2$ in disguise. In linear algebra this is a statement that the quadratic form $(n-1)S-2P$ is positive semi-definite, with the all-ones vector spanning its kernel. Strang's *Introduction to Linear Algebra* covers quadratic forms and positive definiteness, and Siklos's *Advanced Problems in Mathematics* has more STEP-style inequality practice.

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